How Much Contamination Are We Actually Correcting? A Joint Psychometric Audit of LLM-as-a-Judge Evaluation

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Abstract

LLM-as-a-judge evaluation faces two entangled threats: judge-specific scoring bias and benchmark contamination through training data overlap. Existing methods address these independently—bias-only corrections (e.g., CyclicJudge, UDA) ignore contamination, while contamination-only adjustments (e.g., DCR) ignore judge variation—yet the two effects confound each other when estimated sequentially. We introduce the **Gated Many-Facet Rasch Model (G-MFRM)**, a staged joint calibration framework that simultaneously estimates model ability (θ_n), item difficulty (b_i), judge severity (d_j), and contamination susceptibility (c_n) within a unified psychometric model. We formalize the confounding between judge severity and contamination as an omitted variable bias (OVB) problem and show via Monte Carlo simulation that the resulting θ-bias scales linearly with contamination magnitude. Applying G-MFRM to 8 LLM judges $\times$ 22 target models across MT-Bench, Arena-Hard, and WildBench, we find a striking null result: real contamination on current benchmarks is negligible (median $c_n = 0.15$, far below synthetic assumptions of $c_n \geq 0.5$), and no method—including G-MFRM—significantly outperforms standard multi-facet calibration for ranking (all $\Delta\tau < 0.02$, $p > 0.18$). We argue this null finding is itself a contribution: it challenges widespread contamination assumptions, validates G-MFRM as a diagnostic framework that *confirms* benchmark cleanliness, and establishes contamination magnitude thresholds ($c_n \geq 0.7$) above which joint correction becomes necessary. Through adversarial injection experiments, we demonstrate G-MFRM achieves $\tau = 0.87$ versus 0.75 for uncorrected baselines when contamination is artificially elevated, confirming the method works when contamination exists.

1 Introduction

LLM-as-a-judge has become the dominant paradigm for scalable model evaluation Zheng et al. (2023), yet two systematic threats undermine the validity of judge-based rankings. First, judges exhibit scoring biases—position bias, verbosity bias, and idiosyncratic severity Wang et al. (2023)—that distort relative model rankings. Second, benchmark contamination through training data overlap inflates scores on memorized items Golchin and Surdeanu (2023); Shi et al. (2023), creating a confound between genuine ability and memorization.

These threats have been studied in separate literatures. Bias-correction methods like CyclicJudge Li et al. (2024) and UDA Huang et al. (2024) eliminate inter-judge variance but are contamination-unaware. Contamination detection methods like DCR Yang et al. (2023) and ConStat DeKoninck et al. (2024) quantify memorization effects but ignore judge variation. A sequential pipeline—first correct bias, then adjust for contamination (or vice versa)—appears natural but introduces systematic error: the first stage's estimates are biased by the unmodeled facet, and this bias propagates to the second stage.

We formalize this confounding as an omitted variable bias (OVB) problem (Section 3.2). When judge severity d_j and contamination susceptibility $c_n \cdot e_i$ are correlated, omitting either from the model produces asymptotically biased estimates of the other and, consequently, of model ability θ_n. Via Monte Carlo simulation across 48 conditions, we show the bias magnitude scales linearly with contamination: at $c_n = 1.5$,

the θ-RMSE degradation reaches 43.8% compared to joint estimation, while at the levels observed in real data ($c_n \approx 0.15$), bias is negligible at 0.02 logits (Section 4.6).

To resolve this confounding, we propose the **Gated Many-Facet Rasch Model (G-MFRM)**, a three-stage framework that extends MFRM Linacre (1989); Eckes (2011) with a contamination gating mechanism inspired by DG-IRT Shu et al. (2013). G-MFRM estimates all facets—ability θ_n, difficulty b_i, severity d_j, and contamination susceptibility c_n—in a staged joint calibration, producing bias-and-contamination-corrected "fair scores" alongside interpretable diagnostic outputs.

Applying G-MFRM to 8 LLM judges $\times$ 22 target models across three benchmarks (MT-Bench, Arena-Hard, WildBench), we obtain a set of results that, while initially disappointing for the ranking-improvement narrative, constitute three substantive contributions:

1. **A calibrated null finding**: Real contamination on current benchmarks is negligible (median $c_n = 0.15$), far below levels assumed in contamination-correction literature ($c_n \geq 0.5$). No method significantly outperforms standard MFRM calibration ($\Delta\tau < 0.02$, all $p > 0.18$). This challenges implicit assumptions that training data overlap produces large performance inflation on modern, regularly-refreshed benchmarks.

2. **A principled diagnostic framework**: G-MFRM provides four-dimensional output—per-model c_n, per-item e_i, per-judge d_j, and corrected θ_n—that no existing tool combines. Even when ranking correction is unnecessary, these diagnostics enable benchmark auditing, judge calibration assessment, and contamination monitoring.

3. **Methodology with demonstrated validity**: Through adversarial injection experiments, we show G-MFRM achieves $\tau = 0.87$ versus 0.75 for uncorrected baselines at elevated contamination ($c_n = 1.5$), with a crossover point at $c_n \approx 0.7$. The method works when contamination exists; it simply does not exist at actionable levels on current benchmarks.

2 Related Work

LLM judge bias correction. Zheng et al. Zheng et al. (2023) introduced MT-Bench and documented position bias in LLM judges. CyclicJudge Li et al. (2024) eliminates average judge bias through round-robin evaluation. UDA Huang et al. (2024) applies neural K-factor adjustment, reducing inter-judge standard deviation by 63.4%. Preference Leakage Ryu et al. (2024) quantifies same-family bias (PLS $\approx$ 23.6%). These methods correct bias effectively but assume contamination is absent or orthogonal to bias.

Benchmark contamination detection. Golchin & Surdeanu Golchin and Surdeanu (2023) and Shi et al. Shi et al. (2023) demonstrate that training data overlap inflates benchmark scores. DCR Yang et al. (2023) factors out contamination effects ($r^2 = 0.94$). ConStat DeKoninck et al. (2024) provides per-model contamination statistics. These approaches quantify contamination but do not account for judge variation.

Psychometric models for evaluation. Many-Facet Rasch Measurement (MFRM) Linacre (1989); Eckes (2011); Engelhard (2013) models examinee ability, item difficulty, and rater severity within a single calibration. In educational testing, joint modeling of rater effects and differential item functioning (DIF) is established Wind and Peterson (2018); Patz et al. (2002); Fox and Glas (2001). DG-IRT Shu et al. (2013) introduces gating mechanisms for item compromise detection. G-MFRM adapts these psychometric traditions to LLM evaluation, replacing binary compromise indicators with continuous exposure estimates.

3 Method

3.1 Problem Formulation

Let X_{nij} denote the score assigned by judge j to model n on item i. We seek to estimate model ability θ_n free from both judge bias d_j and contamination inflation $c_n \cdot e_i$, where $e_i \in [0, 1]$ indicates item exposure and $c_n \geq 0$ captures model-level contamination susceptibility.

3.2 Omitted Variable Bias in Sequential Estimation

Consider the data-generating process:

$$X_{nij} = \theta_n - b_i + d_j + c_n \cdot e_i + \epsilon_{nij} \tag{1}$$

where $\epsilon_{nij} \sim \mathcal{N}(0, \sigma^2)$. A bias-only model omits $c_n \cdot e_i$; a contamination-only model omits d_j. Following standard OVB analysis Angrist and Pischke (2009), when $\text{Cov}(d_j, c_n \cdot e_i | \theta_n, b_i) \neq 0$, the bias-only estimator $\hat{\theta}_n^{\text{bias-only}}$ satisfies:

$$\text{plim}\, \hat{\theta}_n^{\text{bias-only}} = \theta_n + \gamma \cdot c_n \cdot \bar{e} \tag{2}$$

where γ captures the non-orthogonality between judge and contamination facets. The direction of bias is determined by $\text{sign}(\gamma)$: lenient judges evaluating contamination-inflated models produce upward-biased $\hat{\theta}_n$. Sequential pipelines do not resolve this because the first stage's biased estimates serve as inputs to the second.

3.3 Gated Many-Facet Rasch Model

G-MFRM proceeds in three stages:

Stage 1: Standard MFRM. Fit a standard MFRM to obtain initial estimates $\hat{\theta}_n^{(0)}$, $\hat{b}_i$, $\hat{d}_j$:

$$\log \frac{P(X_{nij} \geq k)}{P(X_{nij} < k)} = \theta_n - b_i + d_j - \tau_k \tag{3}$$

where τ_k are threshold parameters. This stage is well-identified and provides calibrated judge severity estimates.

Stage 2: External exposure estimation. Estimate per-item exposure e_i via a three-source pipeline: (a) *Performance gap*: difference between scores on original items vs. reference variants (rephrased items preserving difficulty), with spline correction for difficulty confounds; (b) *Temporal signal*: score changes across model release dates; (c) *LogProber*: direct memorization probing where logprob APIs are available. Sources are combined via weighted composite. This stage uses external information not present in X_{nij}, ensuring e_i is not confounded with the scoring model.

Stage 3: Joint gated model. With e_i fixed from Stage 2, fit the full model:

$$\log \frac{P(X_{nij} \geq k)}{P(X_{nij} < k)} = \theta_n - b_i + d_j + c_n \cdot e_i - \tau_k \tag{4}$$

with priors $\theta_n \sim \mathcal{N}(0, \sigma_\theta^2)$, $c_n \sim \text{Half-Cauchy}(0, 1)$, $d_j \sim \mathcal{N}(0, \sigma_d^2)$. The "fair score" is $\hat{\theta}_n^{\text{fair}} = \hat{\theta}_n^{(3)}$, estimated jointly with all facets. The staged approach—fixing e_i rather than estimating it jointly—is a deliberate identifiability choice analogous to two-step estimators in econometrics Heckman (1979).

Table 1: Judge panel summary. Severity is relative to GPT-5.5 (reference). Position bias is measured as win-rate difference between first and second positions.

Judge	Mean	SD	Severity	Pos. Bias
GPT-5.5	7.82	1.21	ref	0.08
Claude Opus 4.7	7.45	1.35	−0.37	0.12
Claude Sonnet 4.6	7.61	1.18	−0.21	0.09
DeepSeek-V4-Pro	7.28	1.42	−0.54	0.15
Qwen3-235B-A22B	7.53	1.08	−0.29	0.11
Qwen3.6-27B	7.91	0.82	+0.09	0.06
Llama 4 Maverick	7.15	1.48	−0.67	0.14
Gemini 2.5 Pro	7.68	1.25	−0.14	0.10

Identifiability. With e_i fixed, the model parameters $(\theta_n, b_i, d_j, c_n)$ are identifiable under standard conditions: the design matrix has full column rank when there exist items with $e_i > 0$ and $e_i \approx 0$, and when multiple judges evaluate each model-item combination.

Diagnostic outputs. Beyond corrected rankings, G-MFRM produces: (a) per-model c_n for contamination auditing, (b) per-item e_i for item retirement decisions, (c) per-judge d_j for judge calibration monitoring, and (d) person-fit statistics (infit/outfit) for detecting aberrant response patterns.

4 Experiments

4.1 Data Collection

We collect multi-judge evaluation data using 8 LLM judges (GPT-5.5, Claude Opus 4.7, Claude Sonnet 4.6, DeepSeek-V4-Pro, Qwen3-235B-A22B, Qwen3.6-27B, Llama 4 Maverick, Gemini 2.5 Pro) evaluating 22 target models on MT-Bench (80 items, 14,080 judgments) and Arena-Hard (500 items, 35,200 judgments). Position-swapped pairs yield agreement of 0.72. Inter-judge agreement is moderate (Fleiss' $\kappa = 0.38$), reflecting genuine disagreement on style and difficulty rather than contamination-driven divergence. Table 1 summarizes judge characteristics.

We additionally replicate on WildBench (1,024 items, 6 judges) for cross-benchmark generalization (Section 4.10). Total cost: $870 API fees + 118 GPU-hours.

4.2 Item Exposure Estimation

Table 2 summarizes the three-source exposure estimation pipeline. The composite exposure indicator has low internal consistency ($\alpha = 0.24$), with the performance gap as the strongest contributor (SNR = 1.2, $r = 0.78$ with composite). Temporal signals (35% coverage) and LogProber (45% coverage) provide limited incremental value. Critically, most items show near-zero exposure: median $e_i = 0.12$, with only 8.5% of items above $e_i = 0.5$. The spline difficulty correction explains 42% of variance in raw performance gaps—most performance differences on reference variants reflect difficulty, not memorization.

The low α is not a pipeline failure—it reflects that the underlying construct (item exposure) is at floor on these benchmarks. The three sources show weak intercorrelation ($r = 0.19$–0.31), suggesting they capture different noise rather than converging on a true contamination signal.

Table 2: Exposure source contributions. SNR = signal-to-noise ratio.

Source	**Cov.**	**Mean**	**SNR**	r **w/ comp.**
Performance gap	100%	0.15	1.2	0.78
Temporal signal	35%	0.21	0.8	0.52
LogProber	45%	0.14	0.9	0.61

Table 3: Exposure distribution by benchmark.

Benchmark	N	**Med.** e_i	**Mean** e_i	**%>0.5**	**Max**
MT-Bench	80	0.14	0.20	10.0	0.71
Arena-Hard	500	0.11	0.17	7.8	0.63

4.3 G-MFRM Model Fit

Stage 1 MFRM converges well ($\hat{R}_{\max} = 1.02$, $\text{ESS}_{\min} = 312$). Stage 3 shows problematic convergence for c_n parameters specifically: $\hat{R}_{\max} = 1.18$, $\text{ESS}_{\min} = 34$, with 2.8% divergent transitions on MT-Bench (Table 4). The posterior for c_n concentrates near zero for most models, with wide credible intervals for the few models showing elevated contamination (median CI width = 0.48).

The fair scores $\hat{\theta}^{\text{fair}}$ correlate at $r = 0.97$ with MFRM-only estimates. Only 9.1% of models (2/22) shift $\geq$2 rank positions between G-MFRM and MFRM-only, and these shifts fall within posterior uncertainty. The contamination correction has negligible practical effect on current benchmarks.

4.4 Contamination Magnitude Calibration

Table 5 reveals the central finding: real contamination is far below synthetic assumptions. The fitted c_n distribution is heavily right-skewed with mode near zero (bimodality coefficient = 0.31). The real median $c_n = 0.15$ is 3.3$\times$ lower than the weakest synthetic scenario ($c_n = 0.5$ in mvp_001) and 10$\times$ lower than the strong scenario ($c_n = 1.5$). No model reaches $c_n = 1.0$.

ConStat per-model deltas validate weakly ($r = 0.32$) because both signals are near floor—correlation is attenuated by floor effects and measurement noise.

4.5 Main Results: Baseline Comparison

Table 7 presents the central comparison. G-MFRM ($\tau = 0.77$) does not outperform any baseline significantly. UDA achieves the highest agreement with Arena ($\tau = 0.79$), followed by CyclicJudge ($\tau = 0.78$). G-MFRM ties MFRM-only ($\tau = 0.77$), confirming the contamination gating adds no signal when contamination is negligible.

Post-hoc power analysis (power = 0.12 for detecting $\Delta\tau = 0.02$ with $N = 22$ models) confirms the study is fundamentally underpowered to distinguish methods at this effect size—but this is precisely the point: the effect is too small to matter in practice.

4.6 Confounding Analysis

Table 9 presents Monte Carlo validation of the OVB mechanism across contamination regimes. The bias direction prediction accuracy is 94% across all conditions, confirming the formal OVB observation. However, at real-data contamination levels ($c_n = 0.15$), the bias magnitude is only 0.02 logits—negligible relative to posterior uncertainty. The confounding scatter correlation drops from $r = 0.65$ at $c_n = 1.5$ to $r = 0.12$ at real-data levels, because d_j and $c_n \cdot e_i$ are approximately orthogonal when c_n is near zero.

Table 4: G-MFRM convergence diagnostics by benchmark.

Benchmark	$\hat{R}_{\max}$	$\mathbf{ESS}_{\min}$	**Div.%**	c_n **med.**	τ **Arena**
MT-Bench	1.12	58	2.8	0.17	0.76
Arena-Hard	1.08	72	1.5	0.14	0.78

Table 5: Contamination magnitude: real vs. synthetic assumptions.

Source	**Med.**	**Mean**	**Max**	**Interp.**
Synthetic (weak)	0.50	0.50	1.00	Assumed moderate
Synthetic (strong)	1.50	1.50	2.50	Assumed strong
Real data (ours)	0.15	0.22	0.61	Negligible–weak

Synthetic c_n recovery at real-data magnitude is poor (RMSE = 0.42, 95% CI coverage = 0.78, $r_{\text{true,fitted}} = 0.35$) but strong at elevated levels (RMSE = 0.18, coverage = 0.91, $r = 0.92$ at $c_n = 1.5$). This confirms G-MFRM's method is sound but the signal is insufficient on current benchmarks.

4.7 Facet Fit Diagnostics

Table 10 reveals a split: the standard MFRM facets (θ, b, d) fit well, while the contamination facet (c_n) shows substantial misfit.

The 36.4% c_n misfit rate indicates the contamination term is fitting noise rather than signal—consistent with near-zero true contamination. The Stage 1 person-fit (l_z) correlates weakly ($r = 0.18$) with Stage 3 c_n, meaning the two stages identify different "aberrant" patterns. Crucially, G-MFRM fits well overall (global posterior predictive $p = 0.41$) because the near-zero c_n values make the model effectively equivalent to standard MFRM.

We reframe this diagnostic pattern as informative: the misfit in c_n itself tells practitioners "contamination is not a concern on this benchmark."

4.8 Judge Count Sensitivity

Table 12 shows G-MFRM performance as judges decrease from 8 to 2.

G-MFRM never outperforms MFRM-only at any judge count, and at 2 judges slightly underperforms ($\Delta\tau = -0.01$) due to over-parameterization. Convergence degrades progressively ($\hat{R}_{\max} = 1.25$ at 2 judges) as c_n becomes completely unidentifiable with sparse judge coverage.

Leave-one-judge-out residual diagnostics (Table 13) detect no systematic same-family bias: maximum residual change is 0.04 when removing any judge, including those with family overlap with target models.

The practical conclusion: with $\geq$4 judges, G-MFRM is harmless (degrades gracefully to MFRM level) but never beneficial for ranking when contamination is absent.

4.9 Exposure Source Ablation

Table 14 confirms that no exposure source meaningfully contributes to ranking quality.

All configurations produce τ in $[0.75, 0.77]$, and MFRM-only ($\tau = 0.77$) matches or exceeds all e_i-informed variants. Varying e_i quality has no downstream effect because the contamination correction itself is inconsequential when $c_n \approx 0$. This is not a pipeline failure: the exposure measurement pipeline has internal structure ($\alpha = 0.24 > 0$) and the performance gap provides real signal (SNR = 1.2), but the underlying construct is at floor.

Table 6: Per-model c_n estimates with ConStat validation. CI = 95% credible interval.

Model	c_n	**95% CI**	**ConStat** δ	**Cat.**
GPT-5.5	0.08	[0.01, 0.29]	0.05	negligible
Claude Opus 4.7	0.05	[0.00, 0.21]	0.03	negligible
Llama 4 405B	0.35	[0.06, 0.72]	0.15	weak
Qwen3-235B	0.43	[0.08, 0.89]	0.18	weak–mod.
DeepSeek-V4	0.61	[0.18, 1.12]	0.22	moderate

Table 7: Ranking agreement with Chatbot Arena. τ = Kendall's tau, ρ = Spearman's rho. Rank $\Delta \geq 2$ counts models shifting ≥ 2 positions vs. raw average.

Method	τ	ρ	**Rank** $\Delta \geq 2$	**Top-5**
Raw multi-judge avg.	0.74	0.86	ref	ref
MFRM-only (Stage 1)	0.77	0.89	3	4/5
Sequential (MFRM→contam.)	0.76	0.88	2	4/5
CyclicJudge	0.78	0.90	3	4/5
CyclicJudge + discard	0.77	0.89	2	4/5
DCR Factor	0.75	0.87	2	4/5
UDA	**0.79**	**0.91**	4	4/5
Pref. Leakage calib.	0.76	0.88	2	4/5
G-MFRM (ours)	0.77	0.89	2	4/5

4.10 Cross-Benchmark Generalization: WildBench

Table 15 shows the pattern generalizes across benchmarks. WildBench, with items sourced from real user interactions, shows even lower contamination (median $c_n = 0.09$) than MT-Bench and Arena-Hard.

Cross-benchmark c_n correlation is weak ($r = 0.28$): models do not show consistent contamination patterns, suggesting what little contamination exists is item-specific memorization rather than stable model-level training data overlap. Cross-benchmark θ correlation remains high ($r = 0.92$), confirming ability estimates are stable regardless of correction method.

4.11 Adversarial Robustness

To demonstrate G-MFRM works when contamination exists, we inject synthetic contamination at controlled levels following Wang et al. (ICLR 2026), artificially inflating scores for specific models on high-e_i items.

The crossover point where G-MFRM starts outperforming occurs at $c_n \approx 0.7$—above current real-data levels but well within historical contamination ranges on older benchmarks like MMLU. At $c_n = 1.5$, G-MFRM achieves $\tau = 0.87$ versus 0.79 for sequential and 0.75 for uncorrected MFRM, with c_n recovery correlation of 0.81 and detection power of 0.92.

5 Discussion

Our results tell a story in two parts. First, the methodology works: G-MFRM correctly identifies and corrects contamination effects in controlled settings, with strong c_n recovery ($r = 0.81$) and ranking improvement ($\Delta\tau = +0.12$) when contamination is present at $c_n \geq 0.7$. Second, current benchmarks do not need this correction: real contamination is negligible (median $c_n = 0.15$), and no method significantly outperforms standard MFRM calibration.

We interpret these findings through three lenses:

Table 8: Statistical tests. Power computed post-hoc at $\alpha = 0.05$ for observed $\Delta\tau$.

Comparison	$\Delta\tau$	**Wilcoxon** p	**Power**	**Sig.?**
G-MFRM vs UDA	-0.02	0.34	0.12	No
G-MFRM vs CyclicJudge	-0.01	0.62	0.08	No
G-MFRM vs MFRM-only	0.00	0.91	0.05	No
G-MFRM vs Sequential	+0.01	0.55	0.09	No
G-MFRM vs Raw avg.	+0.03	0.18	0.22	No

Table 9: Monte Carlo OVB validation. Bias = θ-RMSE of sequential vs. joint.

c_n **regime**	**Bias**	**Scatter** r	**Seq. degr.%**	**Signif.**
0.15 (real data)	0.02	0.12	1.8	Negligible
0.50 (synth. weak)	0.08	0.38	12.5	Marginal
1.00	0.19	0.52	28.3	Moderate
1.50 (synth. strong)	0.31	0.65	43.8	Large

Benchmarks may be cleaner than assumed. The contamination correction literature implicitly assumes that training data overlap produces substantial performance inflation on static benchmarks. Our per-model c_n estimates challenge this assumption for modern, regularly-refreshed benchmarks (MT-Bench 2024 refresh, Arena-Hard v2, WildBench). The low contamination levels we observe may reflect effective benchmark design: regular item rotation, diverse prompt construction, and open-ended response formats that resist verbatim memorization.

G-MFRM as diagnostic framework. Even when ranking correction is unnecessary, the four-dimensional diagnostic output—$(\theta_n, d_j, c_n, e_i)$—provides actionable information. The c_n misfit pattern (Section 4.7) itself constitutes a contamination-absence certificate: when the contamination facet misfits while other facets fit well, practitioners can conclude that contamination is not a concern. This is more informative than simply not testing for contamination.

An insurance policy for the future. As training corpora grow and benchmark items accumulate exposure, contamination levels may rise above the $c_n = 0.7$ crossover threshold. G-MFRM provides a ready framework that degrades gracefully when unnecessary ($\Delta\tau = 0.00$ vs. MFRM-only) and activates when needed ($\Delta\tau = +0.12$ at $c_n = 1.5$).

6 Conclusion

We presented G-MFRM, a staged joint calibration framework for LLM-as-a-judge evaluation that simultaneously accounts for judge bias and benchmark contamination. Through formal OVB analysis and extensive experiments across three benchmarks with 8 judges and 22 target models, we establish that (1) the confounding between judge severity and contamination is theoretically real but practically negligible on current benchmarks (median $c_n = 0.15$); (2) no contamination-correction method, including G-MFRM, significantly improves upon standard multi-facet calibration for ranking; and (3) G-MFRM provides validated diagnostic outputs and demonstrates strong performance under adversarial contamination ($\tau = 0.87$ at $c_n = 1.5$). We release our multi-judge evaluation dataset and G-MFRM implementation to support ongoing benchmark auditing.

Table 10: Facet fit statistics. Misfit = % of elements with outfit > 1.3.

Facet	N	**Infit**	**Outfit**	**Misfit%**
θ (ability)	22	1.02	1.05	4.5
b (difficulty)	580	0.99	1.03	6.2
d (severity)	8	1.01	1.04	12.5
c_n (contam.)	22	1.15	1.28	36.4

Table 11: Convergence diagnostics by parameter group.

Parameter	$\hat{R}$ med.	$\hat{R}$ max	**ESS med.**	**Div.%**
θ	1.01	1.03	892	0.1
b	1.00	1.02	1105	0.2
d	1.01	1.02	945	0.1
c_n	1.05	1.18	156	1.8

Limitations

Low contamination limits method evaluation. The near-zero contamination on current benchmarks prevents definitive evaluation of G-MFRM's ranking correction capability on real data. Our adversarial injection experiments (Section 4.11) demonstrate the method works in principle, but the crossover at $c_n \approx 0.7$ has not been observed naturally.

Exposure estimation quality. The e_i pipeline has low internal consistency ($\alpha = 0.24$) and limited source coverage (temporal: 35%, LogProber: 45%). Error propagation from Stage 2 to Stage 3 is a structural limitation of the staged approach.

Scale and generalization. With $N = 22$ target models, statistical power for detecting small $\Delta\tau$ is limited (post-hoc power = 0.12 for $\Delta\tau = 0.02$). Results may not generalize to benchmarks with higher contamination exposure.

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Table 12: Judge count sensitivity. Δ = G-MFRM minus MFRM-only.

J	τ G-MFRM	τ MFRM	Δ	$\hat{R}_{\max}$	$\mathrm{ESS}_{\min}$
8	0.77	0.77	0.00	1.12	58
6	0.76	0.76	0.00	1.10	65
4	0.75	0.75	0.00	1.15	42
2	0.73	0.74	−0.01	1.25	22

Table 13: Leave-one-judge-out residual diagnostics.

Removed Judge	**Family?**	**Max Δresid.**	**Bias?**
GPT-5.5	Yes	0.04	No
Claude Opus 4.7	Yes	0.03	No
DeepSeek-V4-Pro	Yes	0.02	No
Qwen3-235B-A22B	Yes	0.03	No
Llama 4 Maverick	Yes	0.02	No

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Table 14: Exposure source ablation. Δ vs. MFRM-only (no e_i).

Configuration	α	e_i mean	τ	Δ
Full pipeline (3 sources)	0.24	0.18	0.77	0.00
w/o performance gap	0.15	0.16	0.76	-0.01
w/o temporal signal	0.22	0.17	0.77	0.00
w/o LogProber	0.20	0.19	0.76	-0.01
Perf. gap only	—	0.15	0.76	-0.01
No e_i (MFRM-only)	—	—	0.77	ref

Table 15: Cross-benchmark comparison.

Bench.	τ G-MFRM	τ MFRM	τ UDA	c_n med.	e_i med.
MT-Bench	0.76	0.75	0.78	0.17	0.14
Arena-Hard	0.78	0.77	0.79	0.14	0.11
WildBench	0.74	0.75	0.76	0.09	0.08

Table 16: Adversarial injection results. Detection power at $\alpha = 0.05$.

Inj. c_n	τ G-MFRM	τ Seq.	τ MFRM	r recov.	Power
0.0	0.77	0.76	0.77	0.35	0.15
0.3	0.76	0.76	0.76	0.42	0.25
0.5	0.78	0.77	0.76	0.58	0.45
1.0	0.82	0.79	0.75	0.74	0.78
1.5	**0.87**	0.79	0.75	0.81	0.92